

(1001CJA102120031)

Test Pattern

**CLASSROOM CONTACT PROGRAMME**

(Academic Session : 2020 - 2021)

JEE(Advanced)

REVIEW TEST

13-09-2020

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TOAS)**ANSWER KEY****PAPER-1****PART-1 : PHYSICS**

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A,B,D	A,C	A,C	B,C	B,D	B,C	C,D	A,B,C	A,C	A,C
SECTION-II	Q.	1	2	3	4	5	6	7	8		
	A.	0.88 to 0.89	2.40	70.60 to 70.70	1.70	1.50	0.00	0.00	4.50		

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	C,D	A,C,D	B,C,D	A,B,C	B,D	A,C,D	B,C,D	B,C,D	D
SECTION-II	Q.	1	2	3	4	5	6	7	8		
	A.	4.00	4.00	3.00	4.00	5.00	2.00	19.00	6.00		

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A,B,D	B,D	B,D	A,D	B,D	A,B,C	B,D	A,B	A,B,C	C,D
SECTION-II	Q.	1	2	3	4	5	6	7	8		
	A.	3.00	3.00	2.00	0.86 or 0.87	0.20	1.00	3.00	2.00		

(1001CJA102120032)

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JEE(Advanced)

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13-09-2020

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TOAS)**ANSWER KEY****PAPER-2****PART-1 : PHYSICS**

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A,B	A,C,D	A,B	A	A,B,D	A,C,D	B	B	B	A
	Q.	11	12								
	A.	C	B								
SECTION-II	Q.	1	2	3	4	5	6				
	A.	3.92 to 3.93	0.50	49.96 to 50.00	336.00	25.00	0.68				

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A,C	A,B,C,D	B,C,D	A,B,C,D	A,D	A,B	B	C	C	C
	Q.	11	12								
	A.	C	C								
SECTION-II	Q.	1	2	3	4	5	6				
	A.	2.00	8.00	5.00	3.00	20.00	4.00				

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A,C	D	B	A,C,D	A,B	A,C,D	D	D	A	B
	Q.	11	12								
	A.	C	C								
SECTION-II	Q.	1	2	3	4	5	6				
	A.	6.00	0.66 or 0.67	31.00	64.00	11.00	-3.00				



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JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TOAS)

PAPER-1

PART-2 : CHEMISTRY

SOLUTION

SECTION-I

1. Ans.(B)
2. Ans.(C,D)
3. Ans.(A,C,D)
4. Ans.(B,C,D)
5. Ans.(A,B,C)
6. Ans.(B,D)
7. Ans.(A, C, D)

fcc unit cell

packing fraction = 0.74

$$a\sqrt{2} = 4r \text{ octahedral void}$$

$$r_v = 0.414 \times r$$

$$r_v = 0.414 \times \frac{200\sqrt{2}}{4} = 29.27 \text{ pm}$$

distance of nearest neighbours =

$$\frac{a\sqrt{2}}{2} = \frac{200\sqrt{2}}{2} = 141.4 \text{ pm}$$

distance of next neighbours = $a = 200 \text{ pm}$.

8. Ans.(B,C,D)
9. Ans.(B,C,D)
10. Ans.(D)

SECTION-II

1. Ans. (4.00)
2. Ans.(4.00)
 $\Delta T_f = iK_f m$
 $\Rightarrow 3.255 = i \times 1.86 \times 0.35$
 $\Rightarrow i = 5$
3. Ans.(3.00)
4. Ans.(4.00)
5. Ans.(5.00)
6. Ans.(2.00)
7. Ans.(19.00)

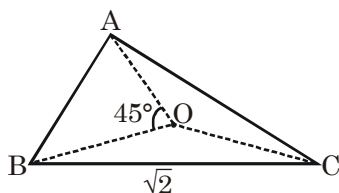
- Sol. (a) = KO_2
 (b) = $\text{K}_2[\text{PtCl}_6]$
 (c) = MnO_4^-
 (d) = O_2
8. Ans.(6.00)

PART-3 : MATHEMATICS

SOLUTION

SECTION-I

1. Ans. (A,B,D)
- Sol. $\therefore \text{OA} + \text{OB} = 4$
 $\Rightarrow \text{OA} \cdot \text{OB} \leq 4$



$$\therefore \text{ar}(\triangle AOB) = \frac{1}{2}(\text{OA})(\text{OB}) \times \frac{1}{\sqrt{2}} \leq \sqrt{2}$$

Let h = altitude drawn from the vertex C to the base AOB.

$$\text{So, } h \leq \sqrt{2}$$

$$\therefore V = \frac{1}{3} \times \text{ar}(\triangle AOB) \times h \leq \frac{1}{3} \times \sqrt{2} \times \sqrt{2} = \frac{2}{3}$$

But $V = \frac{2}{3} \Rightarrow$ equality holds everywhereSo, $\text{OA} = \text{OB} = 2$ & $h = \sqrt{2}$ Also, $\text{BC} \perp \text{base AOB} \Rightarrow \angle \text{OBC} = 90^\circ$

$$\text{So, } \text{OC} = \sqrt{6}$$

$$\text{And, } \text{AB} = \sqrt{4 + 4 - 8 \times \frac{1}{\sqrt{2}}} = \sqrt{8 - 4\sqrt{2}}$$

$$\therefore \text{AC} = \sqrt{8 - 4\sqrt{2}} + 2 = \sqrt{10 - 4\sqrt{2}}$$

$$> \sqrt{9 - 4\sqrt{2}} = (2\sqrt{2} - 1)$$

2. Ans: (B,D)

$$\text{Sol.: } \text{tr}((A - AB)(A^T - B^T A^T)) = 0$$

$$\Rightarrow A - AB = O_n$$

$$\text{Since, } \text{tr}(CC^T) = 0$$

$$\Rightarrow C = O_n$$

3. Ans. (B,D)

$$\text{Sol. } \vec{n}_1 = 2\hat{i} - 2\hat{j} - \hat{k} \Rightarrow \hat{n}_1 = \frac{2\hat{i} - 2\hat{j} - \hat{k}}{3}$$

$$\vec{n}_2 = 8\hat{i} + \hat{j} - 4\hat{k} \Rightarrow \hat{n}_2 = \frac{8\hat{i} + \hat{j} - 4\hat{k}}{9}$$

$$\hat{n}_1 + \hat{n}_2 = \frac{14\hat{i} - 5\hat{j} - 7\hat{k}}{9}, \hat{n}_1 - \hat{n}_2 = \frac{-2\hat{i} - 7\hat{j} + \hat{k}}{9}$$

$\therefore$ Angle bisector of lines are parallel to $(\hat{n}_1 + \hat{n}_2)$ and $(\hat{n}_1 - \hat{n}_2)$

$$\Rightarrow \text{Equation of planes} \Rightarrow \vec{r} \cdot (\hat{n}_1 + \hat{n}_2) = 0$$

$$14x - 5y - 7z = 0 \quad \dots(1)$$

$$\Rightarrow \vec{r} \cdot (\hat{n}_1 - \hat{n}_2) = 0$$

$$2x + 7y - z = 0 \quad \dots(2)$$

4. Ans.(A,D)

Sol. Images of (α, β) in the two tangents are $(\alpha,$

$-\beta)$ and (β, α) . Slope of axis = $\frac{1}{2}$.

$$\text{So, } \frac{-\beta}{\alpha - 1} = \frac{\alpha - 1}{\beta - 1} = \frac{1}{2} \Rightarrow (\alpha, \beta) = \left(\frac{3}{5}, \frac{1}{5}\right)$$

5. Ans. (B,D)

Sol. Points are $(0, 0)$ $(t^2, 2t)$ & $\left(\frac{1}{t^2}, -\frac{2}{t}\right)$

$$h = \frac{t^2 + \frac{1}{t^2}}{3}$$

$$\Rightarrow \left(t - \frac{1}{t}\right)^2 + 2 = 3h$$

$$K = \frac{2\left(t - \frac{1}{t}\right)}{3}$$

$$\Rightarrow t - \frac{1}{t} = \frac{3K}{2}$$

$$\frac{9y^2}{4} + 2 = 3h$$

$$9y^2 = 12h - 8$$

$$y^2 = \frac{12}{9} \left(h - \frac{2}{3}\right)$$

$$y^2 = \frac{4}{3} \left(x - \frac{2}{3}\right)$$

6. Ans. (A,B,C)

Sol. Let P $(at^2, 2at)$

Normal at P passes through centre of circle $y = -tx + 2t + t^3$ passes through $(2, 8)$

$$\Rightarrow t = +2$$

Point P(4,4)

Slope normal = -2

$$\text{Slope of tangent} = \frac{1}{2}$$

7. Ans.(B,D)

$$f(x) = \begin{cases} |x-1|([x]-1+\{x\}) & x \notin \text{int} \\ |x-1|[x] & x \in \text{int} \end{cases}$$

$$f(x) = \begin{cases} (x-1)^2 & x > 1, x \notin \text{int} \\ -(x-1)^2 & x < 1, x \notin \text{int} \\ x(x-1) & x > 1, x \in \text{int} \\ x(1-x) & x < 1, x \in \text{int} \end{cases}$$

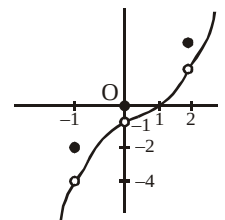
$f(x)$ is continuous at $x = 1$

$f(x)$ can't take value '-1'

Range of $f(x)$ for $x \in [-1, 2]$

is $(-4, 1) \cup \{2\} - \{-1\}$

$$f'(1^+) = 0, f'(1^-) = 0$$



8. Ans. (A,B)

Sol. $f(x) = e^{e^x} - \tan^{-1} x - 3$

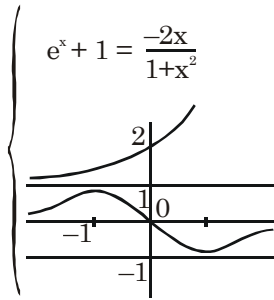
$$f'(x) = e^{e^x} \cdot e^x - \frac{1}{1+x^2}$$

$$f'(x) = \frac{e^{e^x+x}(1+x^2) - 1}{1+x^2}$$

$$\text{Let } g(x) = e^{e^x+x}(1+x^2) - 1$$

$$g'(x) = e^{e^x+x} \cdot \{e^x + 1\} \{1 + x^2\} + e^{e^x+x} \cdot 2x$$

$$g'(x) = e^{e^x+x} \{(e^x + 1)(1 + x^2) + 2x\}$$



Now $(e^x + 1)(1 + x^2) + 2x > 0$

$\therefore g'(x) > 0 \forall x \in \mathbb{R}$

$\lim_{x \rightarrow -\infty} g(x) = -1$ and $g(0) > 0$

$\therefore f(x)$ has negative minima

$\lim_{x \rightarrow -\infty} f(x) < 0$ and $f(0) < 0$

$f(x) = 0$ has positive real root

9. **Ans. (A,B,C)**

$$xdy - y^2dy = ydx \Rightarrow -y^2dy = ydx - xdy$$

$$\Rightarrow -dy = d\left(\frac{x}{y}\right)$$

$$\therefore -y = \frac{x}{y} + c$$

$$-2 = \frac{4}{2} + c \Rightarrow c = -4$$

$$\therefore -y^2 = x - 4y$$

$$x = 4y - y^2 = 4 - (y - 2)^2$$

$$(y - 2)^2 = -(x - 4)$$

$$\therefore \text{focus is } \left(\frac{15}{4}, 2\right)$$

equation of directrix

$$x = \frac{17}{4} \Rightarrow 4x - 17 = 0$$

$$e = 1.$$

10. **Ans. (C,D)**

$$f(x) - e^x + \frac{1}{f(x)} - e^{-x} = 0$$

$$(f(x) - e^x) + \frac{e^x - f(x)}{e^x f(x)} = 0$$

$$(f(x) - e^x)(e^x f(x) - 1) = 0$$

$$\Rightarrow f(x) = e^x \text{ or } e^{-x}$$

$$\text{or } \begin{cases} e^x, & x \geq 0 \\ e^{-x}, & x < 0 \end{cases} \text{ or } \begin{cases} e^x, & x < 0 \\ -e^{-x}, & x \geq 0 \end{cases}$$

$$\int_0^1 e^x dx = e - 1$$

$$\int_0^1 e^{-x} dx = 1 - \frac{1}{e}$$

$$\therefore \text{Area can be } e - 1 + 1 - \frac{1}{e}$$

$$\text{or } e - 1 + e - 1 \text{ or } 1 - \frac{1}{e} + 1 - \frac{1}{e}$$

SECTION-II

1. **Ans. 3.00**

$$\overrightarrow{OA} = \vec{a}, \overrightarrow{OB} = \vec{b} \text{ \& } \overrightarrow{OC} = \vec{c}$$

$$\vec{a} + 2\vec{b} + 3\vec{c} = 0$$

$$\vec{a} = -2\vec{b} - 3\vec{c}$$

now proceed

2. **Ans. 3.00**

$$AX = \lambda X \Rightarrow (A - \lambda I)X = 0$$

$$X \neq 0 \Rightarrow \begin{vmatrix} 4-\lambda & 6 & 6 \\ 1 & 3-\lambda & 2 \\ -1 & -5 & -2-\lambda \end{vmatrix} = 0$$

$$\Rightarrow (\lambda - 1)(\lambda - 2)^2 = 0$$

$$\Rightarrow \lambda = 1, 2$$

3. **Ans. 2.00**

$$ax \sec \theta - by \csc \theta = a^2 - b^2$$

$$ax \sec \theta - by \csc \theta = -(a^2 - b^2)$$

$$L = \frac{2(a^2 - b^2)}{\sqrt{a^2 \sec^2 \theta + b^2 \csc^2 \theta}}$$

$$\therefore a^2 \sec^2 \theta + b^2 \csc^2 \theta \geq (a + b)^2$$

$$L \leq 2(a - b)$$

4. **Ans. 0.86 or 0.87**

$$y = -\frac{2a}{\sqrt{1-a^2}}x + \frac{1}{\sqrt{1-a^2}}$$

compare with $y = mx + c$

$$m = -\frac{2a}{\sqrt{1-a^2}}, c = \frac{1}{\sqrt{1-a^2}}$$

using tangency condition $c^2 = A^2 m^2 + B^2$

$$\frac{1}{1-a^2} = \frac{A^2 \cdot 4a^2}{1-a^2} + B^2$$

$$\Rightarrow 1 = a^2(4A^2 - B^2) + B^2$$

$$\Rightarrow B^2 = 1 \text{ and } 4A^2 - B^2 = 0 \Rightarrow A^2 = \frac{1}{4}$$

$$\therefore A^2 = B^2(1 - e^2) \Rightarrow e = \frac{\sqrt{3}}{2}$$

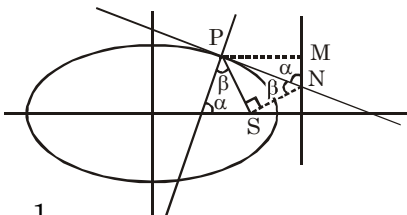
5. Ans. 0.20

$$\sin \alpha = \frac{PM}{PN}$$

$$\sin \beta = \frac{PS}{PN}$$

$$\frac{\sin \beta}{\sin \alpha} = \frac{PS}{PM} = \frac{1}{3}$$

$$\sin \beta = \frac{1}{3} \cdot \sin \alpha = \frac{1}{3} \cdot \frac{3}{5} = \frac{1}{5}$$



6. Ans. 1.00

It is 0° form

$$\text{Let } y = (\sin \theta)^{(\sin \theta - \sin^2 \theta)}$$

$$\ell n y = (1 - \sin \theta)(\sin \theta \ell n \sin \theta)$$

$$\ell n y = \lim_{\theta \rightarrow 0} \frac{\ell n \sin \theta}{\cos \theta} \quad (\text{Use L'Hopital and solve})$$

7. Ans. 3.00

$$\frac{d^3 y}{dx^3} = \frac{d}{dx} \left(\frac{d}{dx} \left(\frac{dy}{dx} \right) \right)$$

$$= \frac{d}{dx} \left(\frac{d}{dy} \left(\frac{1}{\frac{dx}{dy}} \right) \cdot \frac{dy}{dx} \right)$$

$$= \frac{d}{dx} \left(-\frac{1}{\left(\frac{dx}{dy} \right)^2} \cdot \frac{d^2 x}{dy^2} \cdot \frac{dy}{dx} \right)$$

$$= \frac{d}{dx} \left(-\left(\frac{dy}{dx} \right)^3 \cdot \frac{d^2 x}{dy^2} \right)$$

$$= -\left(\frac{dy}{dx} \right)^2 \left(3 \cdot \frac{d^2 x}{dy^2} \cdot \frac{d^2 y}{dx^2} + \left(\frac{dy}{dx} \right)^2 \cdot \frac{d^3 x}{dy^3} \right)$$

8. Ans 2.00

$$F(x) = \int_0^x f(t) dt \Rightarrow xF\left(\frac{1}{x}\right) = x \int_0^{1/x} f(t) dt$$

$$\text{Let } \frac{1}{x} = nT + E$$

$$\lim_{x \rightarrow 0} xF\left(\frac{1}{x}\right) = \lim_{n \rightarrow \infty} \frac{1}{nT + E} \int_0^{nT+E} f(t) dt$$

$$= \lim_{n \rightarrow \infty} \frac{1}{nT + E} \left\{ \int_0^{nT} f(t) dt + \int_{nT}^{nT+E} f(t) dt \right\}$$

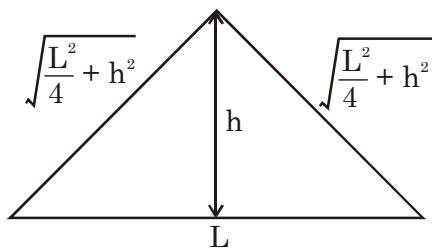
$$= \lim_{n \rightarrow \infty} \frac{n}{nT + E} \int_0^T f(t) dt + \frac{1}{nT + E} \int_0^E f(t) dt$$

$$= \frac{1}{T} \int_0^T f(t) dt = \frac{1}{3} \times 6 = 2$$

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TOAS)**PAPER-2****PART-1 : PHYSICS****SOLUTION****SECTION-I**

1. Ans. (A,B)

Sol. $\lambda = \frac{3 \times 10^8}{3 \times 10^7} = 10$



$$\sqrt{L^2 + 4h_2^2} - \sqrt{L^2 + 4h_1^2} = n\lambda$$

$$L^2 + 4h_2^2 = 100n^2 + L^2 + 4h_1^2 + 20n$$

$$\sqrt{L^2 + 4h_1^2}$$

$$h_2^2 - h_1^2 - 25n^2 = 5n\sqrt{L^2 + 4h_1^2}$$

$$\left(\frac{8 \times 10^{10} - 25n^2}{5n}\right)^2 - 4 \times 10^{10} = L^2$$

2. Ans. (A,C,D)

Sol. Resultant intensity

$$I = I_0 \cos^2\left(\frac{\Delta\phi}{2}\right)$$

$$\Delta x = 400 \text{ nm},$$

$$\Delta\phi = \frac{2\pi}{\lambda} \times \Delta x = \frac{2\pi}{500} \times 400$$

$$\Rightarrow \frac{\Delta\phi}{2} = \frac{4\pi}{5}$$

$$\therefore I_1 = I_0 \cos^2\left(\frac{4\pi}{5}\right)$$

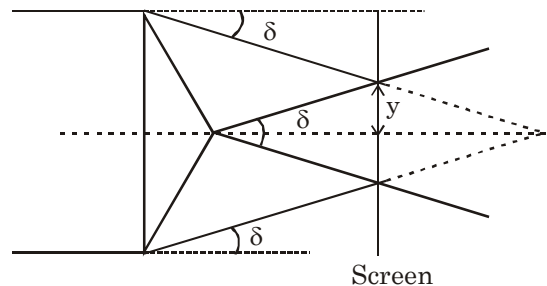
$$\approx 0.64 I_0$$

$$I_2 = I_0 \cos^2\left(\frac{2\pi}{3 \times 2}\right) \Rightarrow \frac{I_0}{4} = 0.25 I_0$$

$$\therefore I_0 > I_1 > I_2$$

3. Ans. (A,B)

Sol.



$$\delta = (n-1)\lambda$$

$$W = \frac{\lambda}{2\delta} = \frac{500}{2 \times 0.5 \times 0.01} = 5 \times 10^{-5} \text{ m}$$

$$y = L \tan \delta$$

$$\frac{D}{2} - y = \tan \delta$$

$$\Rightarrow \frac{D}{2} = 2y$$

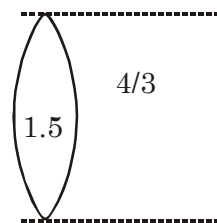
$$y = \frac{D}{4}$$

$$L = \frac{D}{4 \tan \delta} = \frac{6 \text{ cm}}{4 \times 0.01} = 1.5 \text{ m}$$

$$N = \frac{1.5}{5 \times 10^{-5}}$$

4. Ans. (A)

Sol.



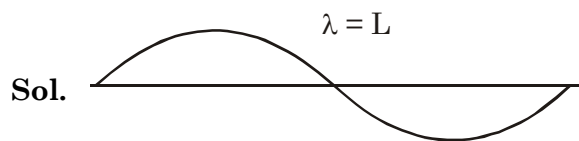
$$\frac{1}{v} - \frac{4}{3u} = \frac{\frac{4}{3} - 1.5}{R} - \frac{1.5 - 1}{-R} = -\frac{1}{6R} + \frac{1}{2R} = \frac{5}{6R}$$

$$\Rightarrow \frac{1}{v} = \frac{5}{6R} - \frac{4}{3u} \rightarrow \text{if } v \text{ is } -ve \text{ he can see.}$$

$$\Rightarrow R - r$$

Only (A)

5. Ans. (A,B,D)



$$y = A \sin kx \sin \omega t$$

$$u_p = \frac{\partial y}{\partial t} = A\omega \sin kx \cos \omega t$$

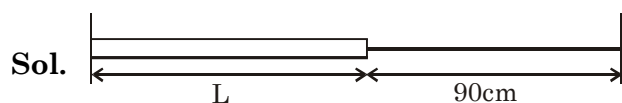
$$(A) \rightarrow A\omega \sin\left(\frac{2\pi}{L} \times \frac{L}{4}\right) \cos\left(\frac{2\pi}{T} \times \frac{T}{3}\right)$$

$$(B) \rightarrow A\omega \sin\left(\frac{2\pi}{L} \times \frac{L}{4}\right) \cos\left(\frac{2\pi}{T} \times \frac{T}{6}\right)$$

$$(C) \rightarrow A\omega \sin\left(\frac{2\pi}{L} \times \frac{L}{3}\right) \cos\left(\frac{2\pi}{T} \times \frac{T}{4}\right)$$

$$(D) \rightarrow A\omega \sin\left(\frac{2\pi}{L} \times \frac{L}{12}\right) \cos\left(\frac{2\pi}{T} \times \frac{T}{2}\right)$$

6. Ans. (A,C,D)



$$f = \frac{n}{2L} \times \sqrt{\frac{T}{\rho \times 4\pi r^2}} = \frac{m}{2 \times 90} \sqrt{\frac{T}{\rho \times \pi r^2}}$$

$$\Rightarrow \frac{n}{m} = \frac{L}{45}$$

$$n + m = 4$$

$$n = 1, 2, 3,$$

$$m = 3, 2, 1$$

$$L = 45 \times \frac{1}{3} = 15\text{cm}$$

$$L = 45 \times 1 = 45\text{cm}$$

$$L = 45 \times \frac{3}{1} = 135\text{cm}$$

7. Ans. (B)

Sol. $f_1 = \sqrt{\frac{c+v}{c-v}} f_0$

$$f_2 = \sqrt{\frac{c+v}{c-v}} f_1$$

$$f_0 + \Delta f = f_2 = \frac{c+v}{c-v} f_0$$

$$\frac{\Delta f}{f_0} = \frac{2v}{c-v} = 3 \times 10^{-7}$$

$$\Rightarrow 2v = 90$$

$$v = 45 \text{ m/s}$$

8. Ans. (B)

Sol. $f = \sqrt{\frac{c+v}{c-v}} f_0$

$$= \sqrt{\frac{1.6}{0.4}} f_0 = 2f_0$$

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{2 \times 10^{15}} = 150\text{nm}$$

9. Ans. (B)

Sol. $(\mu - 1)t = \left(n + \frac{1}{2}\right)\lambda$

$$t = \left(n + \frac{1}{2}\right) \times \frac{560 \times 10^{-9}}{0.4} = \left(n + \frac{1}{2}\right) \times 1.4 \times 10^{-6}$$

Since we observe 3 max. $n \geq 3$

$$\Rightarrow (2n + 1) \times 0.7 \mu\text{m}$$

10. Ans. (A)

Sol. $(\mu - 1) \Delta t = \lambda$ in 30 sec.

$$\frac{\Delta t}{\Delta T} = \frac{560}{0.4} = \frac{140}{3} \text{ nm/s}$$

11. Ans. (C)

Sol. $\frac{3}{2L} \times \sqrt{\frac{T}{\mu}} = \frac{1}{4L} \times v_s$

$$\Rightarrow \sqrt{\frac{T}{\mu}} = \frac{v_s}{60} \Rightarrow T = \frac{\mu v_s^2}{3600}$$

12. Ans. (B)

Sol. $\frac{v_s}{60} \times \frac{5n}{L} = \frac{v_s}{4L} \times (2m - 1)$

$$\Rightarrow \frac{n}{3} = 2m - 1$$

$$\Rightarrow \frac{n+3}{6} = m$$

Put $m = 1, 2, 3, \dots, n = 3, 9, 15, 21, \dots$

SECTION-II

1. Ans. 3.92 to 3.93

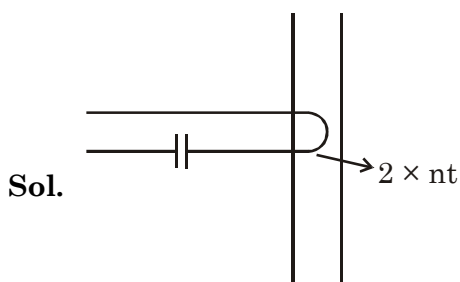
Sol. $\Delta\phi = \frac{2\pi}{\lambda} \Delta x$

$$= \frac{2\pi}{0.40} \times 0.25$$

$$= \frac{5\pi}{4} = 3.92$$

$$\left(\lambda = \frac{340}{850} = 0.4 \right)$$

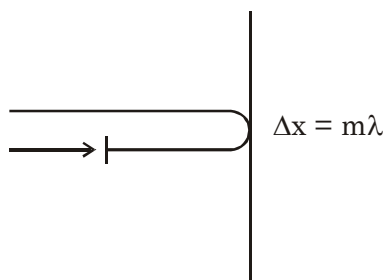
2. Ans. 0.50



$$\Delta x' = \Delta x - 2t + 2nt = \left(m + \frac{1}{2} \right) \lambda$$

$$2(n-1)t = \frac{\lambda}{2}$$

$$t = \frac{\lambda}{4 \times 0.5}$$

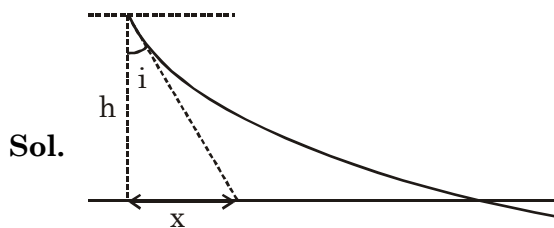


$$\lambda = \frac{3 \times 10^8}{3 \times 10^{11}} = 10^{-3} \text{ m}$$

$$t = \frac{1}{2} \times 10^{-3}$$

$$= 0.5 \text{ mm}$$

3. Ans. 49.96 to 50.00



$$c_0(1 - ah)\sin i = c_0 \sin 90^\circ$$

$$\sin i = \frac{1}{1 - ah} = \frac{x}{\sqrt{h^2 + x^2}}$$

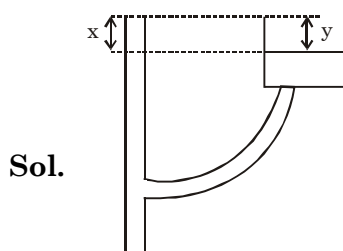
$$x = h \tan i$$

$$= h(1 - ah) \sqrt{1 - \frac{1}{(1 - ah)^2}}$$

$$= \frac{h(1 - ah)}{\sqrt{a^2 h^2 - 2ah}}$$

$$H = \frac{2 \times \left(1 - \frac{2}{2500} \right)}{\sqrt{\frac{4}{2500}}} = 50 \text{ m}$$

4. Ans. 336.00



$$xA = 6A \times (21 - x)$$

$$\frac{7x}{6} = 21$$

$$x = 18 \text{ cm}$$

$$x'A = 6A \times (70 - x)$$

$$\frac{7x'}{6} = 70$$

$$x' = 60$$

$$x' - x = \frac{\lambda}{2} = 42 \text{ cm}$$

$$\lambda = 84 \text{ cm}$$

$$v = 336 \text{ m/s}$$

5. Ans. 25.00

Sol.



$$v = L\sqrt{\frac{g}{8d}} = 1 \times \sqrt{\frac{10}{8 \times 2}} \times 10^3 = \frac{100}{4} = 25 \text{ m/s}$$

$$T \sin \phi = \mu \ell g$$

$$T \times \frac{dy}{d\ell} = \mu \ell g$$

$$T \int dy = \mu g \int_0^{L/2} \ell d\ell$$

$$Td = \frac{\mu g}{2} \left(\frac{L}{2} \right)^2$$

$$T = \frac{\mu g L^2}{8d}$$

$$v = \sqrt{\frac{T}{\mu}} = L\sqrt{\frac{g}{8d}}$$

$$= 25 \text{ m/s}$$

6. Ans. 0.68

Sol. $f = \frac{1}{2\ell} \sqrt{\frac{T}{\mu}}$

$$\Rightarrow \frac{\Delta f}{f} = \frac{1}{2} \times \frac{\Delta T}{T}$$

$$\Rightarrow \frac{\Delta T}{T} = 2 \times \frac{\Delta f}{f}$$

$$\therefore \text{percentage change in } T = 2 \frac{\Delta f}{f} \times 100$$

$$= 2 \times \frac{1.5}{440} \times 100$$

$$= 0.68$$

PART-2 : CHEMISTRY

SOLUTION

SECTION-I

1. Ans.(A,C)
2. Ans.(A,B,C,D)
3. Ans.(B,C,D)
4. Ans.(A,B,C,D)
5. Ans. (A, D)
6. Ans. (A, B)
7. Ans.(B)
8. Ans.(C)
9. Ans.(C)
10. Ans.(C)
11. Ans.(B)

Sol. $e \longrightarrow s$; $\Delta G^0 = -2.76 \text{ kcal/L}$

$$\Delta G^0 = -2.3 RT \log k_{eq}$$

$$-2.76 \times 10^3 = -2.3 \times 2 \times 300 \log k_{eq}$$

$$\log k_{eq} = 2$$

$$k_{eq} = 10^2$$

$$\frac{s}{e} = \frac{100}{1} \Rightarrow \%s = \frac{100}{101} \times 100 = 99\%$$

12. Ans. (C)

Sol. $-2.76 = \Delta G_f^0(s) - \Delta G_f^0(e)$

$$\Rightarrow \Delta G_f^0(e) = -7 + 2.76 = -4.24 \text{ kcal}$$

$$\Delta G^0 = \frac{99}{100} \times \Delta G_f^0(s) + \frac{1}{100} \times \Delta G_f^0(e)$$

$$= \frac{99}{100} \times (-7) + \frac{1}{100} \times -4.24$$

$$= -\frac{693}{100} - \frac{4.240}{100} = -\frac{697.24}{100} \text{ kcal}$$

$$\Delta S^0 = \frac{(-27.9724 + 6.9724)}{300} = \frac{-21 \text{ kcal}}{300} = -70 \text{ cal/k}$$

SECTION-II

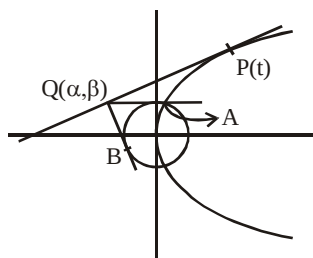
1. Ans.(2.00)
2. Ans.(8.00)
3. Ans. (5.00)
i, ii, iii, vii, viii
4. Ans. (3.00)
5. Ans.(20.00)
6. Ans.(4.00)

PART-3 : MATHEMATICS

SOLUTION

SECTION-I

1. Ans. (A,C)



Equation of chord of contact

$$\alpha x + \beta y = 4 \quad \dots(1)$$

Also, Equation of tangent at P(t)

$$\text{is } ty = x + 2t^2$$

$$\text{Also } t\beta = \alpha + 2t^2 \quad \dots(2)$$

from (1) & (2)

$$(t\beta - 2t^2)x + \beta y = 4$$

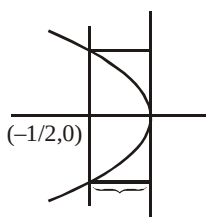
$$\beta(tx + y) - (2t^2x + 4) = 0$$

$$\therefore tx + y = 0 \text{ \& } 2t^2x + 4 = 0$$

$$t = -\frac{y}{x}$$

$$\therefore 2 \cdot \frac{y^2}{x^2} \times x + 4 = 0$$

$$y^2 + 2x = 0$$



Area bounded by S = 0 & the line 2x + 1 = 0

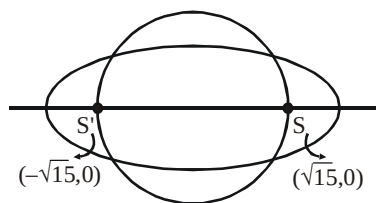
$$\frac{2}{3} \times \frac{1}{2} \times 2 = \frac{2}{3}$$

2x - y + 1 = 0 is a focal chord.

Hence angle is 90°.

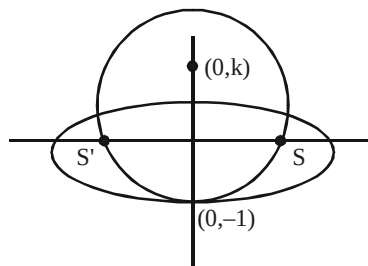
2. Ans. (D)

Least value of $r = \sqrt{15}$



for maximum value of r take limiting case

$$k + 1 = \sqrt{k^2 + 15}$$



$$k = 7$$

$$r = 8$$

$$\text{so } r \in [\sqrt{15}, 8)$$

3. Ans. (B)

$$\text{Let } x^2 e^{-x} = t \Rightarrow (2xe^{-x} - x^2 e^{-x})dx = dt$$

$$\Rightarrow xe^{-x}(2 - x)dx = dt$$

$$\therefore I = \int \frac{dt}{(1+t)^2} = -\frac{1}{1+t} = -\frac{1}{1+x^2 e^{-x}} = \frac{-e^x}{e^x + x^2}$$

4. Ans. (A, C, D)

$$\text{Set of } [\sin^{-1} x] > [\cos^{-1} x] \text{ is } [\sin 1, 1]$$

$$\text{Range of } \left[x + \frac{1}{2} \right] + \left[x - \frac{1}{2} \right] + 2[-x] \text{ is } \{-1, -3\} \text{ as}$$

$$2I + \left[f + \frac{1}{2} \right] + \left[f - \frac{1}{2} \right] + 2(-I + [-f]) = \begin{cases} -3, & f \in \left[0, \frac{1}{2} \right) \\ -1, & f \in \left[\frac{1}{2}, 1 \right) \end{cases}$$

$$\text{Range of } \left[\frac{1}{[x-3]} \right] \text{ is } \{-1, 0, 1\}$$

5. Ans. (A,B,)

$$\text{It is possible } |\vec{a} \times \vec{b}| = 1 = |\vec{c} \times \vec{d}| \text{ \& } \vec{a} \times \vec{b} \parallel \vec{c} \times \vec{d}$$

$$\therefore \vec{a} \perp \vec{b}, \vec{c} \perp \vec{d} \text{ \& } (\vec{a} \times \vec{b}) \parallel (\vec{c} \times \vec{d})$$

$$\vec{a} \cdot (\vec{b} \times \vec{c}) = (\vec{a} \times \vec{b}) \cdot \vec{c} = k(\vec{c} \times \vec{d}) \cdot \vec{c} = 0$$

$$\Rightarrow [\vec{a} \vec{b} \vec{c}] = 0$$

6. Ans. (A,C,D)

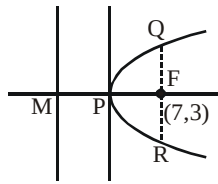
$$(A) \text{ Number of location} = {}^5C_3$$

$$(C) \text{ Number of location} = {}^{10}C_3$$

$$(D) \text{ Number of location} = {}^{10}C_3 - {}^6C_3$$

Paragraph for Question 7 to 8

Reflection of M in the tangent is focus (7,3) as PM is $\perp$ to tangent therefore P is vertex and M foot of directrix
 $(\alpha, \beta) \equiv (5, 1) \Rightarrow \alpha - \beta = 4$
 pair of tangents from M to parabola meet



it at extremity of LR $\frac{x-7}{-\frac{1}{\sqrt{2}}} = \frac{y-3}{\frac{1}{\sqrt{2}}} = \pm 4\sqrt{2}$

$$\Rightarrow (3, 7) (11, -1)$$

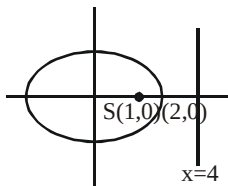
7. **Ans. (D)**

8. **Ans. (D)**

Paragraph for Question 9 to 10

9. **Ans. (A)**

S(1,0) L : x = 4



$$(x-1)^2 + y^2 = (x-4)^2$$

$$y^2 = -6x + 15 \Rightarrow y^2 = -6\left(x - \frac{5}{2}\right)$$

solving C and E

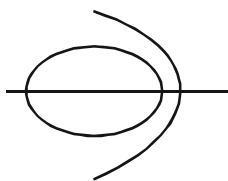
$$\frac{x^2}{4} - \frac{6\left(x - \frac{5}{2}\right)}{3} = 1$$

$$\Rightarrow x^2 - 8\left(x - \frac{5}{2}\right) = 4$$

$$x^2 - 8x + 16 = 0 \Rightarrow x = 4$$

rejected

$\therefore$ No common tangent



Aliter :

For ellipse $PS < PM$
 & For parabola $PS = PM$
 $\Rightarrow$ No common tangent

10. **Ans. (B)**

Let normal of is

$$y = m\left(x - \frac{5}{2}\right) + 3m + \frac{3}{2}m^3$$

$$\text{for tangent } \left(\frac{3}{2}m^3 + \frac{1}{2}m\right)^2 = 4m^2 + 3$$

$$(3m^2 + 1)^2 = 16 + \frac{12}{m^2} \Rightarrow 2 \text{ solution}$$

Paragraph for Question 11 and 12

11. **Ans. (C)**

$$f(x) + f(1-x) = \frac{1+3x^2-3x}{3} + \frac{-1+2x-2x^2}{2} + 1 + \frac{1}{6}$$

$$= \frac{2+6x^2-6x-3+6x-6x^2}{6} + 1 + \frac{1}{6}$$

$$= -\frac{1}{6} + 1 + \frac{1}{6}$$

$$\therefore f(x) + f(1-x) = 1$$

$$\therefore f\left(\frac{1}{100}\right) + f\left(\frac{2}{100}\right) + \dots + f\left(\frac{99}{100}\right)$$

$$= f\left(\frac{1}{100}\right) + f\left(\frac{99}{100}\right) + \dots + f\left(\frac{49}{100}\right) + f\left(\frac{51}{100}\right) + f\left(\frac{1}{2}\right)$$

$$= 1 \times 49 + \frac{1}{24} - \frac{1}{8} + \frac{1}{2} + \frac{1}{12} = \frac{99}{2}$$

12. **Ans. (C)**

$$I = \int_{1/7}^{6/7} f(f(x)) dx = \int_{1/7}^{6/7} f(f(1-x)) dx$$

$$I = \int_{1/7}^{6/7} f(1-f(x)) = \int_{1/7}^{6/7} 1 - f(f(x)) dx$$

$$\therefore 2I = \frac{5}{7}$$

$$\Rightarrow I = \frac{5}{14}$$

SECTION-II

1. Ans. 6.00

$$f(x) = x^4 + px^2 + qx + r \begin{cases} \alpha \\ \alpha \\ \alpha \\ \beta \end{cases}$$

$$f'(x) = 4x^3 + 2px + q \begin{cases} \alpha \\ \alpha \\ \gamma \end{cases}$$

$$f''(x) = 12x^2 + 2p \begin{cases} \alpha \\ \delta \end{cases} \Rightarrow f''(x) = 0$$

$$\Rightarrow \alpha^2 = -\frac{p}{6}$$

Now

$$(1) \alpha^4 + p\alpha^2 + q\alpha + r = 0$$

$$(2) 4\alpha^3 + 2p\alpha + q = 0$$

$$(1) - (2) \times \alpha$$

$$\Rightarrow -3\alpha^4 - p\alpha^2 + r = 0$$

$$-\frac{3p^2}{36} + \frac{p^2}{6} + r = 0$$

$$\Rightarrow p^2 + 12r = 0$$

2. Ans. 0.66 or 0.67

$$\begin{bmatrix} \vec{a} & \vec{b} & \vec{c} \end{bmatrix}^2 = \begin{vmatrix} \hat{a} \cdot \hat{a} & \hat{a} \cdot \hat{b} & \hat{a} \cdot \hat{c} \\ \hat{b} \cdot \hat{a} & \hat{b} \cdot \hat{b} & \hat{b} \cdot \hat{c} \\ \hat{c} \cdot \hat{a} & \hat{c} \cdot \hat{b} & \hat{c} \cdot \hat{c} \end{vmatrix}$$

$$\begin{bmatrix} \vec{a} & \vec{b} & \vec{c} \end{bmatrix}^2 = \begin{vmatrix} 1 & \cos\theta & \cos\theta \\ \cos\theta & 1 & \cos\theta \\ \cos\theta & \cos\theta & 1 \end{vmatrix}$$

$$\begin{bmatrix} \vec{a} & \vec{b} & \vec{c} \end{bmatrix}^2 = (1 - \cos\theta)^2(1 + 2\cos\theta) \geq 0$$

$$\cos\theta \geq -\frac{1}{2}$$

$$\theta = \frac{2\pi}{3}$$

3. Ans. 31.00

$$\therefore BA = AB^2$$

Premultiply by B, we get

$$B^2A = (BA)B^2 = (AB^2)B^2 = AB^4$$

Again premultiply by B

$$B^3A = (BA)B^4 = (AB^2)B^4$$

$$\Rightarrow B^3A = AB^6 \text{ In general } B^m A = AB^{2m} \dots (1)$$

again $BA = AB^2$ post multiply by A^4

$$\Rightarrow BA^5 = AB^2 A^4 \Rightarrow B = AB^2 A^4$$

$$\therefore B = A(B^2A)A^3 = A(AB^4)A^3 \text{ (using (1))}$$

$$\therefore B = A^2(B^4A)A^2 = A^2(AB^8)A^2$$

$$\Rightarrow B = A^3(B^8A)A = A^3(AB^{16})A$$

$$\Rightarrow B = A^4(B^{16}A) = A^4 \cdot AB^{32} \Rightarrow B = B^{32}$$

Pre-multiply by B^{-1} , we get $B^{31} = I$

4. Ans. 64.00

$$A = \begin{bmatrix} \sqrt{3} & 1 & 0 \\ 1 & -\sqrt{3} & 0 \\ 0 & 0 & 2 \end{bmatrix}, A^T = \begin{bmatrix} \sqrt{3} & 1 & 0 \\ 1 & -\sqrt{3} & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

$$\Rightarrow AA^T = 4I$$

$$\Rightarrow A^T = 4A^{-1} = \frac{4 \cdot \text{adj}A}{|A|} = \frac{4 \cdot \text{adj}A}{-8}$$

$$\Rightarrow 2A^T + \text{adj}A = O$$

$$\therefore d = |2A^T + \text{adj}A + AA^T|$$

$$\Rightarrow d = |4I| = 4^3 |I| = 64$$

5. Ans. 11.00

$$f(2x + y^2) = f(2x) + f(y^2) + 10x^2y^2 + 5xy^4 - 4$$

replace $2x \Rightarrow x$ and $y^2 \Rightarrow h$

$$f(x + h) = f(x) + f(h) + \frac{5}{2}x^2h + \frac{5}{2}xh^2 - 4$$

$$\Rightarrow \frac{f(x+h) - f(x)}{h} = \frac{f(h) - 4}{h} + \frac{5}{2}x^2 + \frac{5}{2}xh$$

$$\Rightarrow f'(x) = f'(0) + \frac{5}{2}x^2$$

$$\Rightarrow \frac{dy}{dx} = K + \frac{5}{2}x^2$$

$$\Rightarrow y = Kx + \frac{5x^3}{6} + C$$

$$\text{at } x = 0 \Rightarrow y = 4 \Rightarrow C = 4$$

$$\Rightarrow y = Kx + \frac{5x^3}{6} + 4$$

given $f(1) = 5$

$$\Rightarrow 5 = K + \frac{5}{6} + 4$$

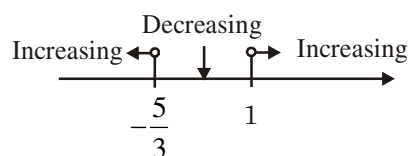
$$\Rightarrow K = \frac{1}{6}$$

$$\text{so } f(x) = \frac{5x^3}{6} + \frac{x}{6} + 4$$

6. Ans. -3.00

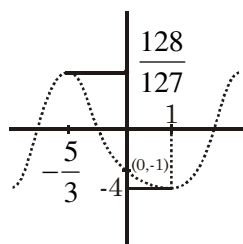
Sol. $f(x) = x^3 + x^2 - 5x - 1$

$$f'(x) = 3x^2 + 2x - 5 = (x - 1)(3x + 5)$$



$$x \rightarrow \infty : f(x) \rightarrow \infty$$

$$x \rightarrow -\infty : f(x) \rightarrow -\infty$$



$$\therefore f(1) = -4$$

$$f\left(\frac{-5}{3}\right) = \frac{128}{127};$$

$$f(10) = -1 < 0$$

$$\left. \begin{array}{l} f(1) = -4 < 0 \\ f(2) = 1 > 0 \end{array} \right\} \text{one root lies between 1 \& 2}$$

$$\left. \begin{array}{l} f(-1) = 4 > 0 \\ f(0) = -1 < 0 \end{array} \right\} \text{one root lies between 0 \& -1}$$

$$\left. \begin{array}{l} f(-2) = 5 > 0 \\ f(-3) = -4 < 0 \end{array} \right\} \text{one root lies between -2 \& -3}$$

$$\therefore [\alpha] + [\beta] + [\gamma]$$

$$= [\text{value between 1 \& 2}] + [\text{value between 0 \& -1}] + [\text{value between -2 \& -3}]$$

$$= 1 - 1 - 3 = -3$$